

Vikram Govindarajan

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PROFESSIONAL LINKS

Website: vikramgovindarajan.github.io
Google Scholar: scholar.google.com
GitHub: github.com/VikramGovindarajan
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SUMMARY

Nuclear engineer specializing in thermal-hydraulics, reactor safety, and scientific computing. Experienced in developing in-house simulation tools for fast reactor system dynamics. Strong interest in adopting state-of-the-art techniques to improve work efficiency, notably, implementing high-performance computing and physics-informed machine learning for transient analysis, undertaking PLM initiative for data and workflow management in the organization, in particular, simulation data and process management.

EDUCATION

Homi Bhabha National Institute (HBNI), Mumbai <i>Master of Technology in Mechanical Engineering, Specialization in Fast Breeder Reactor and Recycle Science & Technology</i>	Kalpakkam, TN Aug. 2014 – Dec. 2016
College of Engineering, Guindy (CEG) <i>Bachelor of Engineering in Mechanical Engineering</i>	Chennai, TN Jul. 2010 – Apr. 2014

EXPERIENCE

Scientific Officer <i>Indira Gandhi Centre for Atomic Research (IGCAR)</i>	Aug. 2014 – Present Kalpakkam, TN
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- Performed plant dynamics and safety analysis for indigenous Fast Breeder Reactors (FBRs)
- Conducted one-dimensional thermal-hydraulic analysis for sodium-cooled reactor systems
- Developed Severe Accident Management Guidelines (SAMG) for fast reactors, a first-of-its kind globally for an operating reactor
- Contributed to IAEA Coordinated Research Projects (CRPs) on reactor safety - FFTF, S-Allegro
- Guided students and trainees in simulation and scientific computing

PROGRAMMING/SOFTWARE SKILLS

Languages: Python, C/C++, Fortran, MATLAB
Software: Flownex
Developer Tools: Git, Sphinx
Libraries: NumPy, Scipy, Matplotlib, CoolProp

PROJECTS

PINET: Thermal-Fluid Simulation Framework

- Developing an in-house general-purpose plant dynamics code for reactor system analysis
- Implementing parallel solvers and domain decomposition techniques using OpenMP, MPI
- Exploring machine learning approaches (PINNs) for transient prediction

SELECTED PUBLICATIONS

Journal articles - 9; Conference papers - 10

- G. Vikram, K. Natesan, M. Rajendrakumar, S.P. Pathak, "Development and validation of transient thermal-hydraulic model for SFR steam generators," *Annals of Nuclear Energy*, 180, 109490
- G. Vikram, A. Riyas, A. Bachchan, M. Rajendrakumar, K. Devan, K. Natesan, "Modeling of sodium-cooled fast reactor core – A sensitivity study on number of core channels and axial meshes," *Nuclear Engineering and Design*, 403, 112136

LANGUAGES

Tamil: Native

English: Professional